

Two-Stage Water Rocket: From AI Simulation to Real-World Experiment

Transcultural & transectoral collaboration in transdisciplinary education

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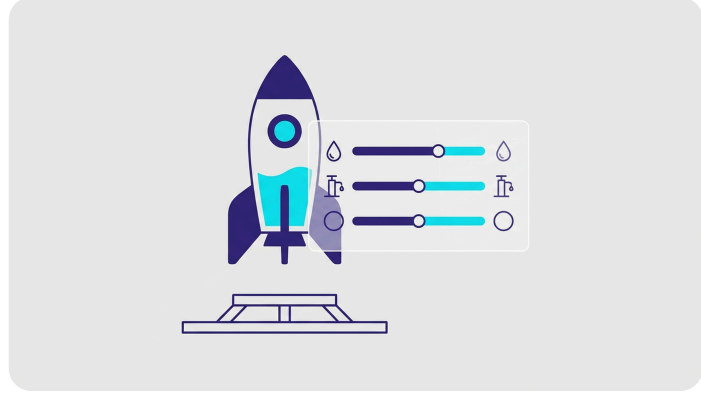
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Background

Teaching abstract mechanics to young children is hard. We designed a bilingual (Chinese–English) STEM unit for primary-school students that pairs an **AI simulation platform** with a **real-world build-and-launch experiment** — delivered as a cross-regional service-learning project by Hong Kong university students.

Transdisciplinary physics × engineering × AI × pedagogy **Transectoral** university ↔ school ↔ community **Transcultural** Hong Kong ↔ Jiangxi, bilingual



Objectives

Knowledge

Explain launch principles, pressure vs. air pressure, action–reaction, acceleration and air resistance.

Ability

Practise **hypothesis** → **test** → **optimise**; variable control; data recording; engineering design.

Transdisciplinary

Couple **simulation** with **hands-on experiment**, transferable to other STEM topics.

Approach & Methodology

1

Learn

Rocket basics · two-stage structure · pressure

2

Build

Rocket assembly · variable control

3

Simulate

AI digital-twin parameter sweep

4

Launch

Formal launch · measure & record

5

Reflect

Simulation vs. real · optimise

The **AI simulation platform** acts as a **pre-launch “digital twin”**: students sweep **water volume**, **air pressure** and **nozzle diameter** to predict the optimal launch *before* the physical test — reducing cost and safety risk while grounding intuition in theory.



Materials



Building



Hands-on



Key Findings & Implications

- Simulation deepened understanding.** The digital twin made abstract forces visible — students saw how changing **water volume**, **air pressure** or **nozzle diameter** changed the launch in real time. This grounded action–reaction and pressure in tangible cause-and-effect, so **[X]**% of learners could explain the underlying mechanism *before* launching.
- Simulation informed the real experiment.** The platform's predicted optimal parameters directly guided the physical launches: students applied the simulated settings and brought the observed trajectory closer to the predicted one across **[M]** launches, reducing trial-and-error.
- Impact & next steps.** Learners gain mechanics intuition and inquiry skills; the **school & community** gain a reusable, low-cost STEM module. Next: map physics one-to-one onto the build/launch steps and add a student innovation zone to close the loop **learn** → **practise** → **create**.

“The children connected pressure and force faster than any worksheet ever did.”

— Teacher testimony (to be confirmed)

“I learned why my rocket flew higher when I pumped in more air.”

— Student testimony (to be confirmed)



References & Acknowledgement

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Scan to explore the full project

github.com/tesolchina/paste2026-poster

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